

25 Equilibria

A · 2026 examination · Concept guide and original exam-style practice

Understand the concept

Weak acid dissociation uses $K_a = \frac{[H^+][A^-]}{[HA]}$, while $K_w = [H^+][OH^-]$. A buffer contains a weak acid and conjugate base: added H^+ is consumed by A^- and added OH^- by HA . For acid buffer, $pH = pK_a + \log([A^-]/[HA])$ when conditions justify approximation.

Worked example: follow the reasoning

For equimolar ethanoic acid and ethanoate, $pH \approx pK_a$ because $\log(1) = 0$.

Quick check · multiple choice

A For equal $[HA]$ and $[A^-]$ in an acid buffer, pH approximately equals what? [1]

- A pK_a
- B K_w
- C 0
- D 14 always

Write your choice: _____

Structured practice

Answer in the space provided. Show working and units where relevant.

1 An acid buffer has $pK_a = 4.76$ and $[A^-]/[HA] = 10$. Determine pH and explain effect of adding a little HCl. [3]

Final answer: _____

Answers and explanation

Quick check: A. pKa

Question 1 [3 marks]

$\text{pH} = 4.76 + \log_{10}(10) = 5.76$ [2]; A^- consumes much of added H^+ , reducing pH change [1].

Self-assessment

Compare each credited step with your own answer. Fix the first reasoning or calculation error; then retry without looking at the solution.

Syllabus: Chemistry 9701, 2025–2027. <https://www.cambridgeinternational.org/Images/664563-2025-2027-syllabus.pdf>

Original FagusED teaching material. Question layout and mark presentation informed by the May/June 2026 papers supplied by the user. The practice question is newly written and is not a Cambridge past-paper question.